# Vector addition



**antocciphysics | TEACHER KEY | CPA + MATCHING SEI**

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**Learning target:** Add displacement vectors and use a model to explain where an object ends up.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

**01. Warm-up: bring back an earlier idea** *Review: Unit 1, Day 3.*
Speed 2 m/s; velocity arrow points east. Speed supplies magnitude only; velocity also supplies direction.
**Look for:** Look for units and an explicit direction, not an unlabeled arrow.

**02. Read and explain a worked model**
Start (0, 0), turn (3, 0), finish (3, 4). Distance 7 m; displacement 5 m northeast. The resultant joins the original start to the final endpoint.
**Look for:** If the student says 7 m northeast, ask which line is being measured.

**03. Finish the model**
Route (0, 0) to (4, 0) to (4, 3); R ends at (4, 3). Components 4 m east, 3 m north. Distance 7 m; displacement 5 m northeast.
**Look for:** Second arrow must start at the first tip; no rotation or rescaling.

**04. A trip that doubles back**
Distance 10 m. Displacement 2 m east. The second leg cancels 4 m of eastward displacement but still adds 4 m of path length.
**Look for:** A negative result may reveal subtracting larger from smaller without first choosing a direction.

**05. Use a new scale**
Distance 14 m; displacement 10 m northeast. Components 6 m east and 8 m north. R is the diagonal from the original start.
**Look for:** Exact angle and use of the Pythagorean theorem are not required in CPA.

**06. Add two diagonal arrows**
Route (0, 0) to (5, 2) to (2, 3). Net 2 m east and 3 m north; northeast.
**Look for:** Check B travels left and upward from A's tip. Do not require a magnitude calculation.

**07. Predict, draw, check**
East-west components cancel. Route distance 10 m; displacement 4 m north.
**Look for:** If the student reports zero displacement, ask what happened to the northward leg.

**08. Find and repair the mistake**
Sam connects the two tips; that arrow is not the combined start-to-finish displacement. Correct route: (0, 0) to (4, 0) to (4, 3); R = (4, 3) m.
**Look for:** A valid repair may also use a correctly constructed parallelogram. Accept an explanation of the endpoints.

**09. One trip, three questions** *Review: Unit 1, Day 2.*
(a) Path: 14 m. (b) Endpoints: 10 m northeast. (c) Path and total time: 14 m / 14 s = 1 m/s.
**Look for:** Do not use 10 m to calculate average speed. That quantity is relevant to average-velocity magnitude.

**10. Exit check - work independently**
Route (0, 0) to (4, 1) to (3, 4). Net 3 m east, 4 m north; northeast. Check endpoint or compare signed component sums with the sketch.
**Look for:** Optional magnitude 5 m is correct but not required by this prompt.

**11. STRETCH: Design two routes**
One route: 4 m east, 3 m north (7 m). Another: 6 m east, 3 m north, 2 m west (11 m). Both end at (4, 3); both have displacement 5 m northeast.
**Look for:** Many routes work. Require both endpoint and total-length checks.

**12. STRETCH: Explore the limits**
Minimum 0 m for opposite directions; maximum 10 m for the same direction. A perpendicular pair gives about 7.1 m; many intermediate examples work.
**Look for:** Opposite equal vectors return to the start. Intermediate magnitude estimates should agree with the drawn scale.

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**01. Retrieve and distinguish** *Review: Unit 1, Day 3.*
Path 6 + 4 = 10 m in 5 s: average speed 2 m/s. Displacement +2 m: average velocity +0.40 m/s (east).
**Look for:** Distance counts both legs; displacement accounts for the westward return.

**02. Explain a worked model**
R = (3, 4) m, 5 m at 53.1° north of east. B runs from (4, 1) to (3, 4), so its x component is negative.
**Look for:** A correct angle is steeper than 45° because the north component exceeds the east component.

**03. Complete, then check**
R = (2, 5) m; √29 = 5.39 m at 68.2° north of east.
**Look for:** Accept 21.8° east of north with a correctly labeled reference.

**04. A different quadrant**
R = (-4,-3) m; 5 m at 36.9° south of west, or 216.9° counterclockwise from +x.
**Look for:** An isolated $\tan^{-1}(3/4)$ value is insufficient without its quadrant.

**05. Choose the correct component signs**
B = (-3.00, 5.196) m. R = (5.660, 10.196) m; 11.66 m at 60.96° north of east.
**Look for:** Angle from west makes the westward cosine component negative. Use degree mode.

**06. The angle is measured from north**
First vector = (5.071, 10.876) m. R = (-1.929, 10.876) m; 11.05 m at 10.06° west of north.
**Look for:** The north component is adjacent to the given 25° angle, so it uses cosine.

**07. Plan the direct return**
Outward path distance 15 m. Final position (6, 5) m. Direct return (-6,-5) m; √61 = 7.81 m at 39.8° south of west.
**Look for:** Do not report the outward resultant direction as the return direction.

**08. Diagnose the reasoning**
The two original vectors are not perpendicular. R = (15, 8.660) m; √300 = 17.32 m at 30.0° north of east.
**Look for:** The Pythagorean theorem applies to perpendicular components, not arbitrary vector magnitudes.

**09. Represent one trip in several ways** *Review: Unit 1, Day 2.*
Distance 14 m; displacement (6, 8) m = 10 m at 53.1° north of east. Average speed 2 m/s. Average velocity (6/7, 8/7) m/s, magnitude 10/7 = 1.43 m/s in the displacement direction. x(t) = 2 t for 0-3 s, then 6 m. y(t) = 0 for 0-3 s, then 2(t -3) for 3-7 s.
**Look for:** The graph slopes give component velocities; start/end changes give the displacement components.

**10. Exit check - solve without the worked model**
R = (3, 2) m; √13 = 3.61 m at 33.7° north of east. C = (-3,-2) m. Tip-to-tail route: (0, 0) to (4,-3) to (3, 2) to (0, 0).
**Look for:** Require route closure or both sums equal to zero; a numerical answer alone is weak evidence.

**11. STRETCH: Missing direction**
$B_x$ = -6 m. $B_y$ = +√(100-36) = +8 m. B is 10 m at 53.1° north of west; R is 8 m north.
**Look for:** The negative square-root branch ends south, so it violates the stated requirement.

**12. STRETCH: Decide what is possible**
Allowed magnitude range is 2-8 m. Thus 1 m and 9 m are impossible; 6 m is possible. For included angle θ: $6^2 = 5^2 + 3^2 + 2(5)(3)\cos\theta$, so θ ≈ 86.2°. The tip-to-tail triangle has supplementary interior angle ≈ 93.8°.
**Look for:** Mirrored arrangements give the same magnitude. A scale estimate near 86° is sufficient unless the optional derivation is assigned.